b.) Range

domain of T

codomain of T

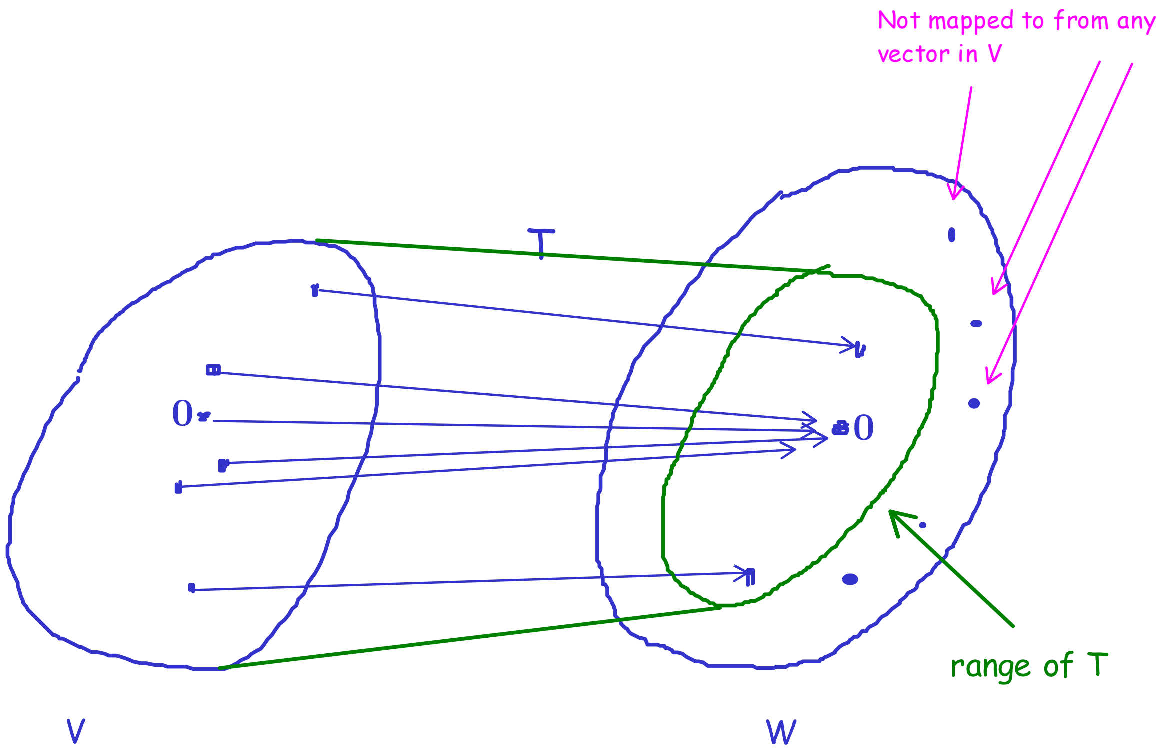
Definition Let $T : V \to W$ be a linear transformation.

Then

$$\mathrm{ran}(T) = \{T(\mathbf{v}) \ : \ \mathbf{v} \in V\}$$

range of T

$$= \{\mathbf{w} \in W \ : \ \mathbf{w} = T(\mathbf{v}) \text{ for some } \mathbf{v} \in V\}$$



Example The linear transformation that sends everything to the zero vector:

$T : V \to W$ such that for all $\mathbf{v} \in V$, $T(\mathbf{v}) = \mathbf{0}$.

Then $\mathrm{ran}(T) = \{\mathbf{0}\}$

Example $T : V \to V$, where for all $\mathbf{v} \in V$, $T(\mathbf{v}) = \mathbf{v}$. (Identity map)

Then $\mathrm{ran}(T) = V$.

Example Let $V = P_3$ and $W = P_2$.

Let $T : P_3 \to P_2$ be defined by

$$T(p(x)) = p'(x).$$

Then

$$\operatorname{ran}(T) = P_2.$$

(every polynomial of degree 2 is the derivative of some function of degree 3 [antiderivative])

m x n matrix

Example $T_A : \mathbb{R}^n \to \mathbb{R}^m$ defined by $T_A(\mathbf{x}) = A\mathbf{x}$.

see for instance Lect. 16 Feb, p 9, "matrix · column vector"

Then

$$\begin{aligned}\operatorname{ran}(T_A) &= \{\mathbf{y} \in \mathbb{R}^m \;:\; \mathbf{y} = A\mathbf{x} \text{ for some } \mathbf{x} \in \mathbb{R}^n\} \\ &= \{\mathbf{y} \in \mathbb{R}^m \;:\; \mathbf{y} \text{ is a linear combination of columns in } A\} \\ &= CS(A)\end{aligned}$$

For instance, if

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$$

Then

$$T_A\left(\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}\right) = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = x_1 \begin{pmatrix} 1 \\ 4 \\ 7 \end{pmatrix} + x_2 \begin{pmatrix} 2 \\ 5 \\ 8 \end{pmatrix} + x_3 \begin{pmatrix} 3 \\ 6 \\ 9 \end{pmatrix}$$

The range is therefore everything that can be written as linear combionation of these three vectors.

Hence the range of T_A is the column space of A.

We have seen that the kernel of T is not just a subset of V, but even a subspace of V.

We now show an analogous statement for the range: The range of T is a subspace of W.

Theorem 1 If $T: V \to W$ is a linear transformation then ran(T) is a subspace of W.

Proof

1.) ran(T) is nonempty because ran(T) always contains the zero vector.

2.) ran(T) is closed under addition:

Let $\mathbf{w}_1$ and $\mathbf{w}_2$ be arbitrary vectors in ran(T).

Then there must exist $\mathbf{v}_1$ and $\mathbf{v}_2$ in V such that

$T(\mathbf{v}_1) = \mathbf{w}_1$ and $T(\mathbf{v}_2) = \mathbf{w}_2$.

Let's determine $T(\mathbf{v}_1 + \mathbf{v}_2)$.

We have

$$T(\underbrace{\mathbf{v}_1 + \mathbf{v}_2}_{\in V}) = T(\mathbf{v}_1) + T(\mathbf{v}_2) = \underbrace{\mathbf{w}_1 + \mathbf{w}_2}_{\text{Hence in the range of T}}$$

Hence $\mathbf{w}_1 + \mathbf{w}_2 \in \operatorname{ran}(T)$.

3.) ran(T) is closed under scalar multiplication:

Let $\mathbf{w}$ be arbitrary vector in ran(T) and λ any scalar.

Then there must exist $\mathbf{v}$ in V such that $T(\mathbf{v}) = \mathbf{w}$.

Let's determine $T(\lambda \mathbf{v})$.

We have

$$T(\underbrace{\lambda \mathbf{v}}_{\in V}) = \lambda T(\mathbf{v}) = \underbrace{\lambda \mathbf{w}}_{\substack{\text{Hence in the} \\ \text{range of T}}}$$

Hence $\lambda \mathbf{w} \in \operatorname{ran}(T)$. $\square$

The following theorem is sometimes useful to determine the range or a basis for the range of a linear transformation.

Theorem 2 Let $T: V \to W$ be a linear transformation.
If $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\}$ spans V then $\{T(\mathbf{v}_1), T(\mathbf{v}_2), \ldots, T(\mathbf{v}_k)\}$ spans ran(T).

Proof Let $\mathbf{w}$ be an arbitrary element of ran(T). Then $\mathbf{w} = T(\mathbf{v})$ for some $\mathbf{v} \in V$. Since $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\}$ spans V, there exist $\lambda_1, \lambda_2, \ldots, \lambda_k$ s.t.

$$\mathbf{v} = \lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \cdots + \lambda_k \mathbf{v}_k.$$

Applying T on both sides of the equation, we get

$$T(\mathbf{v}) = T(\lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \cdots + \lambda_k \mathbf{v}_k),$$

and hence by linearity of T,

$$\mathbf{w} = T(\mathbf{v}) = \lambda_1 T(\mathbf{v}_1) + \lambda_2 T(\mathbf{v}_2) + \cdots + \lambda_k T(\mathbf{v}_k).$$

This tells us that $\mathbf{w}$ is in the span of $\{T(\mathbf{v}_1), T(\mathbf{v}_2), \ldots, T(\mathbf{v}_k)\}$. $\square$

Example Let $T : P_3 \to P_3$ be defined by

$$T(ax^3 + bx^2 + cx + d) = (a-b)x^3 + (c-d)x.$$

We want to find a basis for ran(T).

We apply Thm. 2 to find a set of vectors that spans ran(T).

Let's take any basis in P_3. For instance, we may take the standard basis $B = \{1,\, x,\, x^2,\, x^3\}$

Because B is a basis for P_3, B spans P_3.

Then by Thm. 2, $\{\underset{-x}{\underset{\parallel}{T(1)}},\, \underset{x}{\underset{\parallel}{T(x)}},\, \underset{-x^3}{\underset{\parallel}{T(x^2)}},\, \underset{x^3}{\underset{\parallel}{T(x^3)}}\}$ spans ran(T).

Hence $\{x^3, -x^3, x, -x\}$ spans ran(T).

However, this is obviously not a basis for ran(T) because the vectors are not linearly independent. We can delete (for in instance) $-x^3$ and $-x$ without changing the span. The remaining vectors are linearly independent, and therefore a basis for ran(T).

We therefore have found as a basis for ran(T) the set $\{x^3,\, x\}$.

Remark

Above example that linear transformations in general do not preserve linear independence:

$\{1,\, x,\, x^2,\, x^3\}$ is linearly independent

BUT

$\{T(1),\, T(x),\, T(x^2),\, T(x^3)\}$ is not linearly independent.

However, linear trabsformation do preserve linear dependence!

Proof: Exercise

To be continued